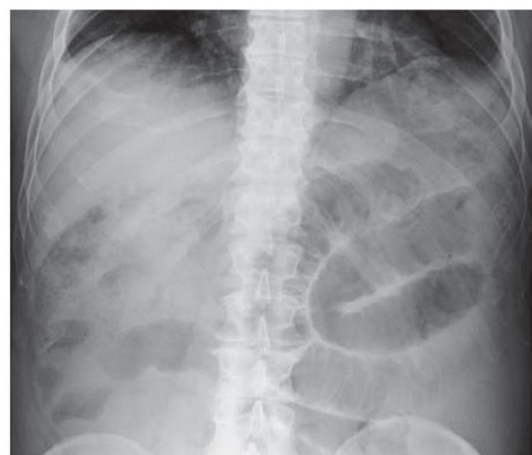




investigations of the GIT system

Imaging : * Plain X-rays

- Plain X-rays of the abdomen are useful in the diagnosis of intestinal obstruction or paralytic ileus, where dilated loops of bowel and (in the erect position) fluid levels may be seen .
- Calcified lymph nodes, gallstones and renal stones can also be detected.
- Chest X-ray (performed with the patient in erect position) is useful in the diagnosis of suspected perforation.

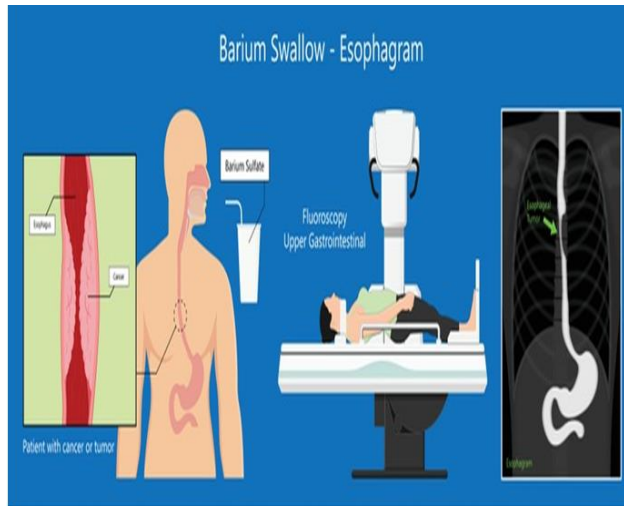


*** Contrast studies**

- X-rays with contrast medium are usually performed to assess not only anatomical abnormalities but also motility. Barium sulphate provides good mucosal coating and excellent opacification but can precipitate impaction proximal to an obstructive lesion.
- Water-soluble contrast is used to opacify bowel prior to abdominal computed tomography and in cases of suspected perforation.
- The double contrast technique improves mucosal visualization by using gas to distend the barium-coated intestinal surface.
- Contrast studies are useful for detecting **filling defects, such as tumors, strictures, ulcers and motility disorders, but are inferior to endoscopic procedures and more sophisticated cross-sectional imaging**



techniques, such as computed tomography and magnetic resonance imaging.



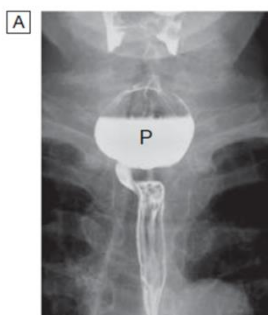
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Double contrast



i 21.3 Contrast radiology in the investigation of gastrointestinal disease			
	Barium swallow/meal	Barium follow-through	Barium enema
Indications and major uses	Motility disorders (achalasia and gastroparesis) Perforation or fistula (non-ionic contrast)	Diarrhoea and abdominal pain of small bowel origin Possible obstruction by strictures Suspected malabsorption Assessment of Crohn's disease	Altered bowel habit Evaluation of strictures or diverticular disease Megacolon Chronic constipation
Limitations	Risk of aspiration Poor mucosal detail Low sensitivity for early cancer Inability to biopsy	Time-consuming nature Radiation exposure Relative insensitivity	Difficulty in frail or incontinent patients Sigmoidoscopy needed to see rectum Low sensitivity for lesions <1 cm





Ultrasound, computed tomography and magnetic resonance imaging

- Ultrasound, computed tomography (CT) and magnetic resonance imaging (MRI) are key tests in the evaluation of intra-abdominal disease. They are non-invasive and offer detailed images of the abdominal contents.
- Fluorodeoxyglucose–positron emission tomography (FDG-PET) is increasingly used in the staging of malignancies and images may be fused with CT to enhance localization.

21.4 Imaging in gastroenterology				
	Ultrasound	Computed tomography (CT)	Magnetic resonance imaging (MRI)	CT–positron emission tomography (CT-PET)
Indications and major uses	Abdominal masses Organomegaly Ascites Biliary tract dilatation Gallstones Guided biopsy of lesions Small bowel imaging	Assessment of pancreatic disease Hepatic tumour deposits CT colonography ('virtual colonoscopy') Tumour staging Assessment of lesion vascularity Abscesses and collections	Hepatic tumour staging MRCP Pelvic/perianal disease Crohn's fistulae Small bowel visualisation	Detection of metastases not seen on ultrasound or CT Images can be fused with CT to form composite image
Limitations	Low sensitivity for small lesions Little functional information Operator-dependent Gas and obesity may obscure view	Cost Radiation dose	Claustrophobic patients Contraindicated in presence of metallic prostheses, cardiac pacemaker, cochlear implants	Signal detection depends on metabolic activity within tumour – not all are metabolically active

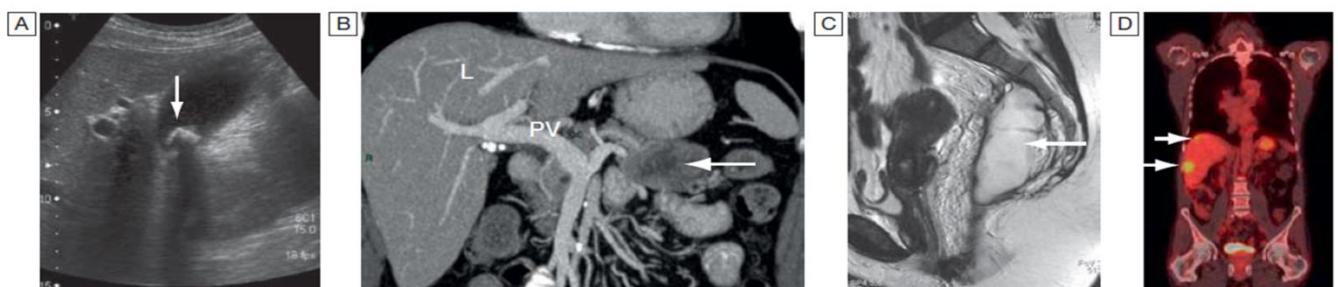


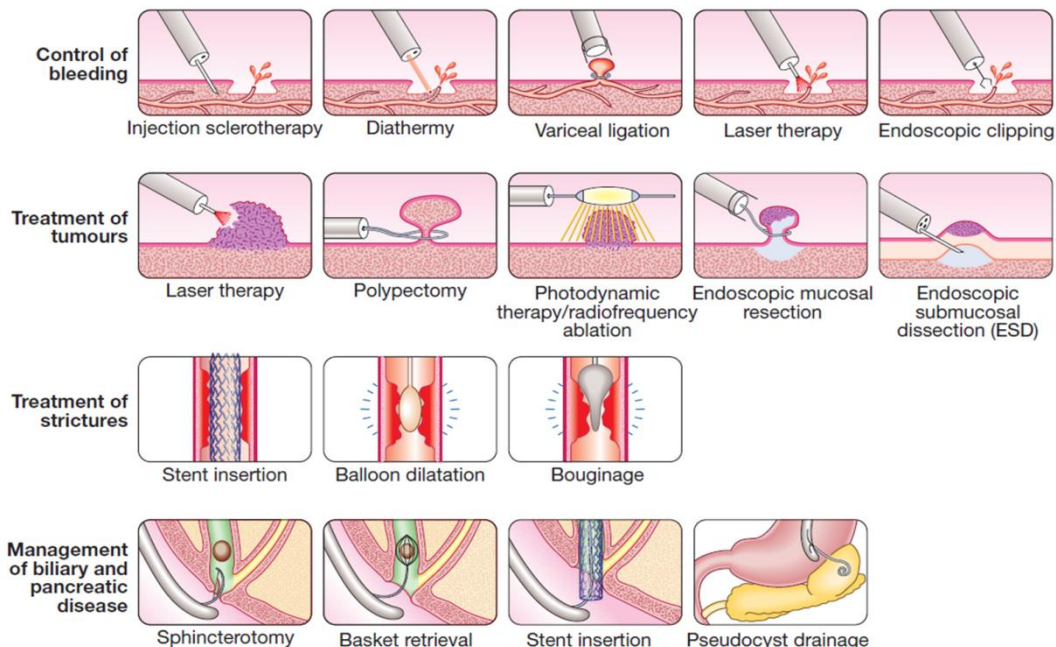
Fig. 21.13 Examples of ultrasound, CT and MRI. **[A]** Ultrasound showing large gallstone (arrow) with acoustic shadowing. **[B]** Multidetector coronal CT showing large solid and cystic malignant tumour in the pancreatic tail (arrow). (PV = portal vein; L = liver) **[C]** Pelvic MRI showing large pelvic abscess (arrow) posterior to the rectum in a patient with Crohn's disease. **[D]** Fused CT-PET image showing two liver metastases (arrows).

Upper gastrointestinal endoscopy

- This is performed under light intravenous benzodiazepine sedation, or using only local anesthetic throat spray after the patient has fasted for at least 6 hours. With the patient in the left lateral position, the entire



esophagus (excluding pharynx), stomach and first two parts of duodenum can be seen.



21.5 Upper gastrointestinal endoscopy	
Indications • Dyspepsia in patients >55 years of age or with alarm symptoms • Atypical chest pain • Dysphagia • Vomiting • Weight loss • Acute or chronic gastrointestinal bleeding • Screening for oesophageal varices in chronic liver disease • Abnormal CT scan or barium meal • Duodenal biopsies in the investigation of malabsorption and confirmation of a diagnosis of coeliac disease prior to commencement of gluten-free diet • Therapy, including treatment of bleeding lesions, banding/injection of varices, dilatation of strictures, insertion of stents, placement of	percutaneous gastrostomies, ablation of Barrett's oesophagus and resection of high-grade dysplastic lesions and early neoplasia in the upper gastrointestinal tract Contraindications • Severe shock • Recent myocardial infarction, unstable angina, cardiac arrhythmia* • Severe respiratory disease* • Atlantoaxial subluxation* • Possible visceral perforation Complications • Cardiorespiratory depression due to sedation • Aspiration pneumonia • Perforation

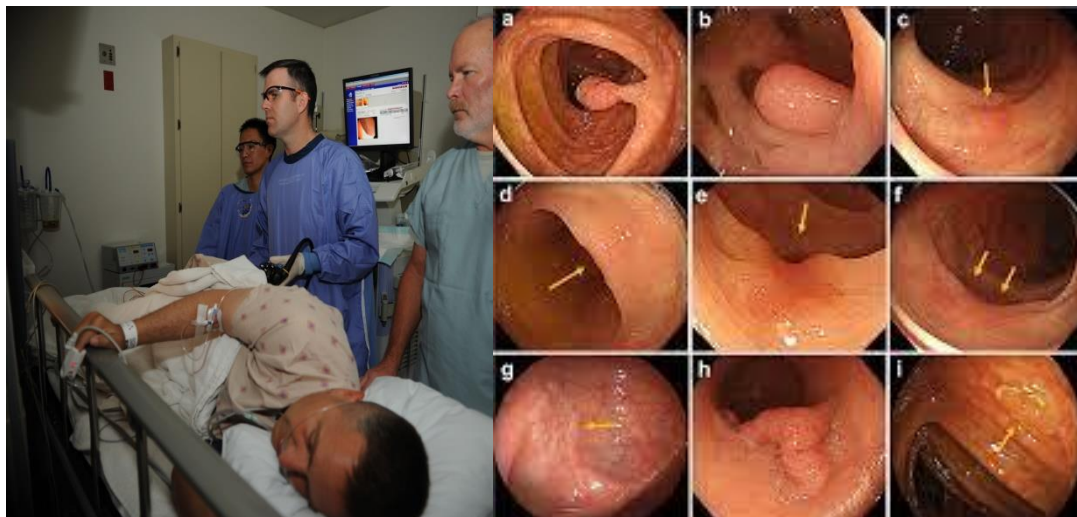
*These are 'relative' contraindications; in experienced hands, endoscopy can be safely performed.



Sigmoidoscopy and colonoscopy

- Sigmoidoscopy can be carried out either in the outpatient clinic using a 20 cm rigid plastic sigmoidoscope or in the endoscopy suite using a 60 cm flexible colonoscope following bowel preparation.
- When sigmoidoscopy is combined with proctoscopy, accurate detection of haemorrhoids, ulcerative colitis and distal colorectal neoplasia is possible.
- After full bowel cleansing, it is possible to examine the entire colon and the terminal ileum using a longer colonoscope.

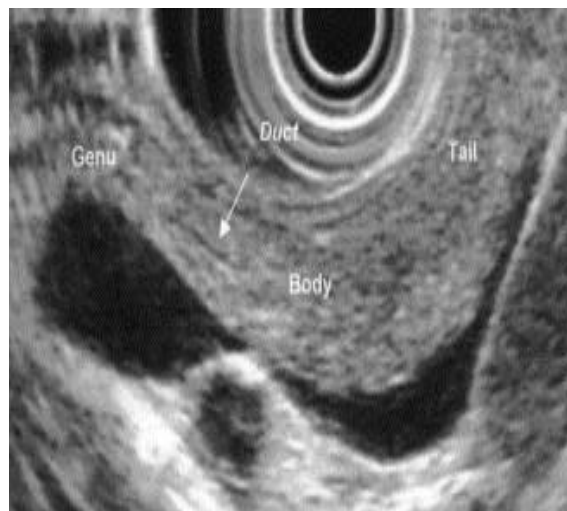
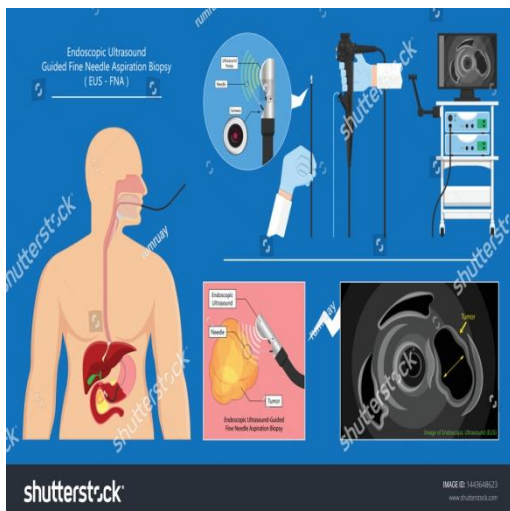
i	21.8 Colonoscopy
	Indications*
	• Suspected inflammatory bowel disease • Chronic diarrhoea • Altered bowel habit • Rectal bleeding or iron deficiency anaemia • Assessment of abnormal CT colonogram or barium enema • Colorectal cancer screening • Colorectal adenoma and carcinoma follow-up • Therapeutic procedures, including endoscopic resection, dilatation of strictures, laser, stent insertion and argon plasma coagulation
	Contraindications
	• Acute severe ulcerative colitis (unprepared flexible sigmoidoscopy is preferred) • As for upper gastrointestinal endoscopy
	Complications
	• Cardiorespiratory depression due to sedation • Perforation • Bleeding following polypectomy
	*Colonoscopy is not useful in the investigation of constipation.





Endoscopic ultrasound

- Endoscopic ultrasound (EUS) combines endoscopy with intraluminal ultrasonography using a high-frequency transducer to produce high-resolution ultrasound images. This allows visualization through the wall of the gastrointestinal tract and into surrounding tissues, e.g the pancreas or lymph nodes.
- It can therefore be used to perform fine needle aspiration or biopsy of mass lesions. EUS is helpful in the diagnosis of pancreatic tumors, chronic pancreatitis, pancreatic cysts, cholangiocarcinoma, common bile duct stones, ampullary lesions and submucosal tumors.
- It also plays an important role in the staging of certain cancers, e.g. those of oesophagus and pancreas. EUS can also be therapeutic, as in drainage of pancreatic fluid collections and coeliac plexus block for pain management. Possible complications of EUS include bleeding, infection, cardiopulmonary events and perforation.

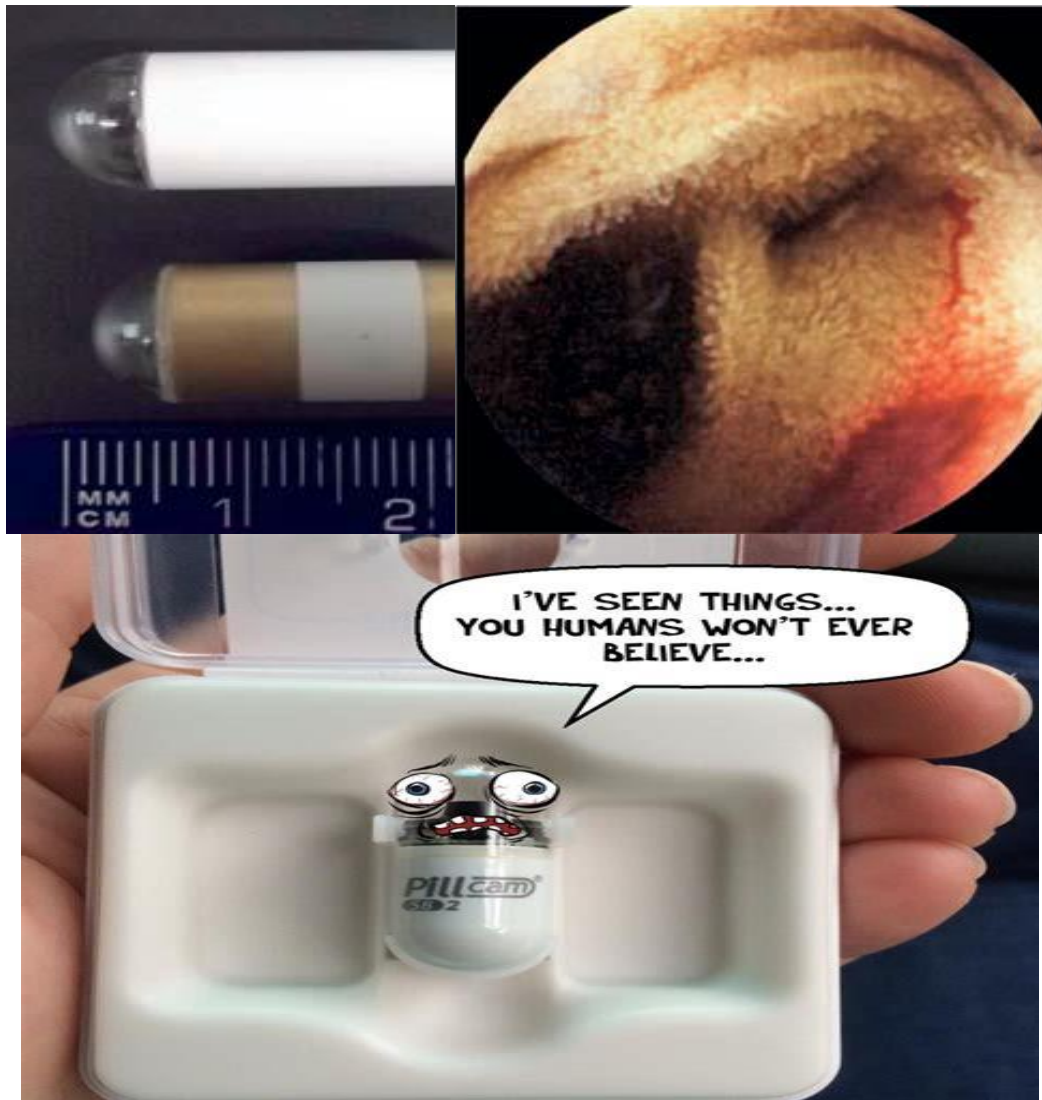


Capsule endoscopy

- Capsule endoscopy uses a capsule containing an imaging device, battery, transmitter and antenna; as it traverses the small intestine, it transmits images to a battery powered recorder worn on a belt round the patient's waist. After approximately 8 hours, the capsule is



excreted. Images from the capsule are analysed as a video sequence and it is usually possible to localise the segment of small bowel in which lesions are seen. Abnormalities detected usually require enteroscopy for confirmation and therapy.

**i****21.6 Wireless capsule endoscopy****Indications**

- Obscure gastrointestinal bleeding
- Small bowel Crohn's disease
- Assessment of coeliac disease and its complications
- Screening and surveillance in familial polyposis syndromes

Contraindications

- Known or suspected small bowel stricture (risk of capsule retention)
- Caution in people with pacemakers or implantable defibrillators

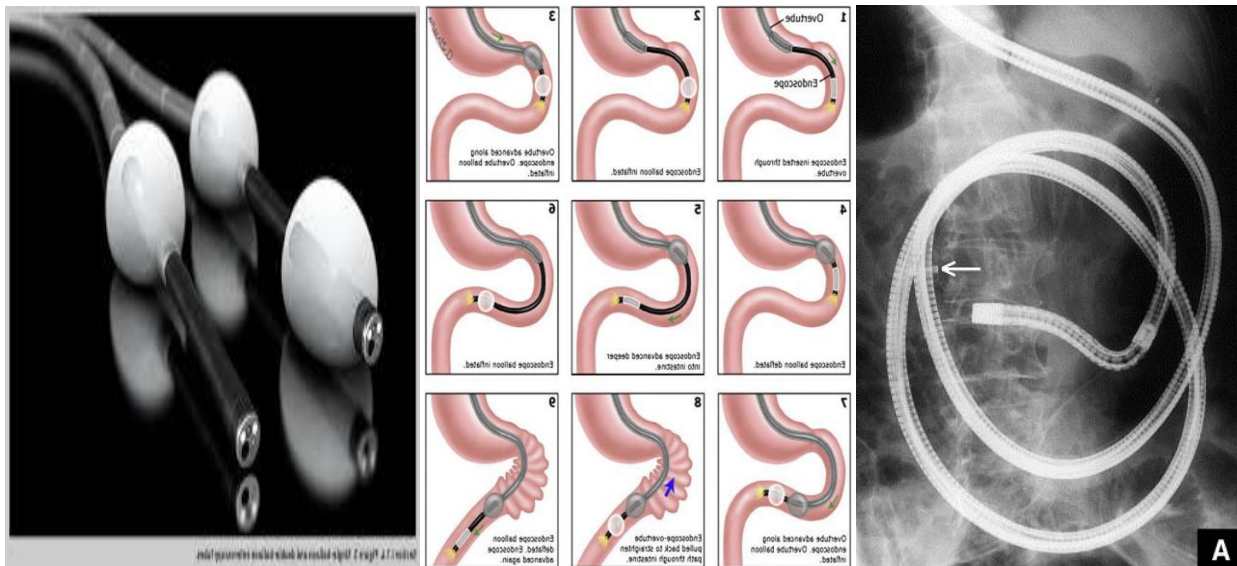
Complications

- Capsule retention (< 1%)



Double balloon enteroscopy

- While endoscopy can reach the proximal small intestine in most patients, a technique called double balloon enteroscopy is also available, which uses a long endoscope with a flexible overtube.
- Sequential and repeated inflation and deflation of balloons on the tip of the overtube and enteroscope allow the operator to push and pull along the entire length of the small intestine to the terminal ileum, in order to diagnose or treat small bowel lesions detected by capsule endoscopy or other imaging modalities.



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21.7 Double balloon enteroscopy

Indications

Diagnostic

- Obscure gastrointestinal bleeding
- Malabsorption or unexplained diarrhoea
- Suspicious radiological findings
- Suspected small bowel tumour
- Surveillance of polyposis syndromes

Therapeutic

- Coagulation/diathermy of bleeding lesions
- Jejunostomy placement

Contraindications

- As for upper gastrointestinal endoscopy

Complications

- As for upper gastrointestinal endoscopy
- Post-procedure abdominal pain ($\leq 20\%$)
- Pancreatitis (1–3%)
- Perforation (especially after resection of large polyps)



Magnetic resonance cholangiopancreatography

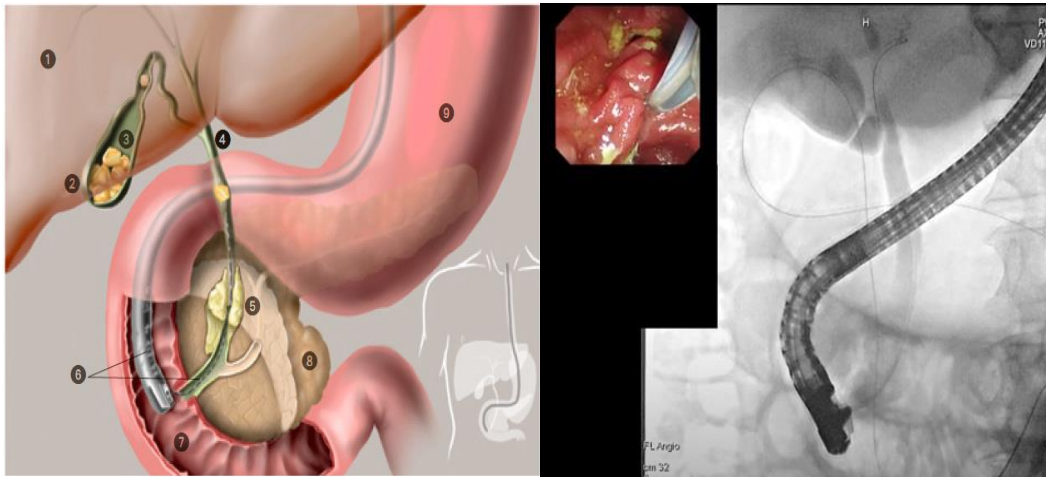
- Magnetic resonance cholangiopancreatography (MRCP) has largely replaced endoscopic retrograde cholangiopancreatography (ERCP) in the evaluation of obstructive jaundice since it produces comparable images of the biliary tree and pancreas, providing information that complements that obtained from CT and endoscopic ultrasound examination (EUS).

Endoscopic retrograde cholangiopancreatography

- Using a side-viewing duodenoscope, it is possible to cannulate the main pancreatic duct and common bile duct. Nowadays, ERCP is used mainly in the treatment of a range of biliary and pancreatic diseases that have been identified by other imaging techniques such as MRCP, EUS and CT.

i	21.10 Endoscopic retrograde cholangiopancreatography
Indications	
Diagnostic	
• Biliary or pancreatic disease where other imaging is equivocal or contraindicated • Ampullary biopsy or biliary cytology	
Therapeutic	
• Biliary disease: - Removal of common bile duct calculi* - Palliation of malignant biliary obstruction - Management of biliary leaks/damage complicating surgery - Dilatation of benign strictures - Primary sclerosing cholangitis • Pancreatic disease: - Drainage of pancreatic pseudocysts and fistulae - Removal of pancreatic calculi (selected cases)	
Contraindications	
• Severe cardiopulmonary comorbidity • Coagulopathy	
Complications	
• Occur in 5–10% with a 30-day mortality of 0.5–1%	
General	
• As for upper endoscopy	
Specific	
• Biliary disease: - Bleeding following sphincterotomy - Cholangitis (if biliary obstruction is not relieved by ERCP) - Gallstone impaction • Pancreatic disease: - Acute pancreatitis - Infection of pseudocyst	
*Laparoscopic surgery is preferred in fit individuals who also require cholecystectomy.	





Histology

- Biopsy material obtained endoscopically or percutaneously can provide useful information

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21.11 Reasons for biopsy or cytological examination

- Suspected malignant lesions
- Assessment of mucosal abnormalities
- Diagnosis of infection (*Candida*, *Helicobacter pylori*, *Giardia lamblia*)
- Analysis of genetic mutations

Serology

- Detection of antibodies plays a limited role in the diagnosis of gastrointestinal infection caused by organisms such as *Helicobacter pylori*, *Salmonella* species and *Entamoeba histolytica*.

Breath tests

- Non-invasive breath tests for *H. pylori* infection and breath tests for suspected small intestinal bacterial overgrowth



Tests of function

- A number of dynamic tests can be used to investigate aspects of gut function, including digestion, absorption, inflammation and epithelial permeability.
- In the assessment of suspected malabsorption, blood tests (full blood count, erythrocyte sedimentation rate (ESR), and measurement of C-reactive protein (CRP), folate, vitamin B12, iron status, albumin, calcium and phosphate) are essential, and endoscopy is undertaken to obtain mucosal biopsies.

21.12 Tests of gastrointestinal function			
Process	Test	Principle	Comments
Absorption	Lactose H ₂ breath test	Measurement of breath H ₂ content after 50 g oral lactose. Undigested sugar is metabolised by colonic bacteria in hypolactasia and expired hydrogen is measured	Non-invasive and accurate. May provoke pain and diarrhoea in sufferers
	⁷⁵ SeHCAT test	Isotopic quantification of 7-day whole-body retention of oral dose ⁷⁵ Se-labelled homocholyltaurine (> 15% = normal, 5–15% borderline, < 5% = abnormal)	Accurate and specific but requires two visits and involves radiation. Results can be equivocal. Serum 7α-hydroxycholestenone is almost as sensitive and specific
	Serum 7α-hydroxycholestenone	Intermediate metabolite of the bile acid synthetic pathway. Serum levels indicate activity of the pathway and are elevated in bile acid diarrhoea	Simple test to perform and only marginally less sensitive and specific than ⁷⁵ SeHCAT test
Pancreatic exocrine function	Pancreolauryl test	Pancreatic esterases cleave fluorescein dilaurylate after oral ingestion. Fluorescein is absorbed and quantified in urine	Accurate and avoids duodenal intubation Takes 2 days. Accurate urine collection essential. Rarely performed
	Faecal elastase	Immunoassay of pancreatic enzymes on stool sample	Simple, quick and avoids urine collection. Does not detect mild disease
Mucosal inflammation/permeability	Faecal calprotectin	A protein secreted non-specifically by neutrophils into the colon in response to inflammation or neoplasia	Useful screening test for gastrointestinal inflammation and for monitoring patients with Crohn's disease and ulcerative colitis. Poor sensitivity for cancer

(⁷⁵SeHCAT = ⁷⁵Se-homocholic acid taurine)



Esophageal motility

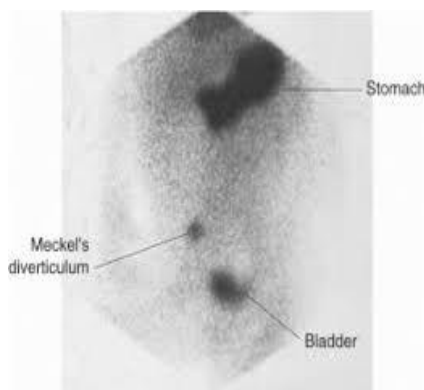
- ✚ A barium swallow can give useful information about esophageal motility.
- ✚ Videofluoroscopy, may be necessary in difficult cases.
- ✚ Oesophageal manometry often in conjunction with 24-hour pH measurements, is of value in diagnosing cases of refractory gastro-esophageal reflux, achalasia and non-cardiac chest pain.

Gastric emptying

- ✓ This involves administering a test meal containing solids and liquids labelled with different radioisotopes and measuring the amount retained in the stomach afterwards . It is useful in the investigation of suspected delayed gastric emptying (gastroparesis) when other studies are normal.

Radioisotope tests

- In some, structural information is obtained, such as the localisation of a Meckel's diverticulum.
- Others provide functional information, such as the rate of gastric emptying or ability to reabsorb bile acids.
- Yet others are tests of infection and rely on the presence of bacteria to hydrolyse a radio-labelled test substance followed by detection of the radioisotope in expired air, such as the urea breath test for *H. pylori*.



21.13 Commonly used radioisotope tests in gastroenterology		
Test	Isotope	Major uses and principle of test
Gastric emptying study	^{99m} Tc-sulphur ¹¹¹ In-DTPA	Assessment of gastric emptying, particularly for possible gastroparesis
Urea breath test	¹³ C-urea	Non-invasive diagnosis of <i>Helicobacter pylori</i> . Bacterial urease enzyme splits urea to ammonia and CO ₂ , which is detected in expired air
Meckel's scan	^{99m} Tc-pertechnetate	Diagnosis of Meckel's diverticulum in cases of obscure gastrointestinal bleeding. Isotope is injected intravenously and localises in ectopic parietal mucosa within diverticulum
Somatostatin receptor scintigraphy (SRS)	¹¹¹ In-DTPA-octreotide	Labelled somatostatin analogue binds to cell surface somatostatin receptors on pancreatic neuro-endocrine tumours
Positron emission tomography (PET)	¹⁸ F-fluorodeoxyglucose (FDG) ⁶⁸ Gallium-labelled somatostatin analogue	Staging high-grade cancers More sensitive and specific than SRS for staging neuro-endocrine tumours

**Gut hormone testing**

- ❖ Excess gut hormone secretion by some gastrointestinal and pancreatic neuro-endocrine tumors can be assessed by measuring levels in blood. Commonly measured hormones include gastrin, somatostatin, vasoactive intestinal polypeptide (VIP) and pancreatic polypeptide.